

Suvankar Payra

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Portfolio: <https://myportfolio.vercel.app>

TECHNICAL SKILLS

Programming: Python, C++, SQL

Backend Development: Flask, REST APIs, Flask-CORS, Jinja2

Python Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

Machine Learning: Regression, Classification, Decision Trees, Random Forest, KNN, SVM, Model Evaluation, Feature Engineering

Databases: MySQL, SQLite

Tools: Git, GitHub, Jupyter Notebook, VS Code

Deployment: Vercel, Render, Streamlit

Core Concepts: OOP, Data Structures & Algorithms, API Development, Exception Handling,

EDUCATION

B.Tech in Computer Science & Engineering

Kazi Nazrul University, Asansol

2024 – 2028 | CGPA: 7.8/10

PROJECTS

Bank Loan Approval Prediction System

([GitHub](#))

- Built a **Python-based full-stack ML web application** that predicts loan approval/rejection from applicant information, with a responsive HTML/CSS/JavaScript interface.
- Developed a **Flask backend and REST API** with a `/predict` POST endpoint to receive applicant data, process it through the prediction pipeline, and return the loan decision.
- Implemented a **modular ML pipeline** covering data cleaning, feature engineering, model training, model serialization, and prediction using Scikit-learn, Pandas, and NumPy.
- Added **application logging and a `/health` endpoint** for debugging and basic backend monitoring, and deployed the complete frontend + backend application on **Vercel**.

FraudShield AI | End-to-End SMS Spam Detection System

([GitHub](#) | [Live Demo](#))

- Built an **end-to-end NLP pipeline** for SMS spam detection, covering **data preprocessing, feature engineering, model training, evaluation, and deployment**.
- Applied **NLTK preprocessing, Word2Vec embeddings, and SMOTE resampling** to improve text classification performance on imbalanced data.
- Trained and compared multiple machine learning models (**Logistic Regression, Random Forest, Gradient Boosting, AdaBoost, Decision Tree, KNN**) using **Precision, Recall, and F1-score metrics**.
- **Developed and deployed a Streamlit web application** for real-time SMS classification with a modular, production-style architecture.

SafeLink Inspector | Phishing Detection System

([GitHub](#) | [Live Demo](#))

- Built a **machine learning-powered phishing detection system** that analyzes URLs and webpage content to classify websites as legitimate or phishing.
- Developed **automated feature extraction pipelines using URL and webpage characteristics, generating 30+ security-related features** for prediction.
- Designed and **deployed an interactive Streamlit web application** for real-time website threat analysis and phishing detection.
- Implemented a modular architecture with model integration, logging, exception handling, and validation workflows to improve maintainability and reliability.

CERTIFICATIONS

- **Complete Machine Learning & Data Science Bootcamp** — Udemy | 2025–2026
- **SQL for Data Analysis** — Simplilearn | 2026